

HACKATHON

Problem Statements

ARTIFICIAL INTELLIGENCE

Theme: AI for Business Intelligence & Automation

The Code Conjurer - Natural Language to Code Snippets

Objective:

Build an AI-powered assistant that converts plain English descriptions into executable code snippets in multiple programming languages.

Features:

- User-friendly UI with a prompt box for natural language input.
- Language selection dropdown (Python, JavaScript, Java, etc.).
- Display generated code with syntax highlighting.
- Export code or copy to clipboard.

Key Parameters:

- Input: Natural language description
- Output: Code snippet (Python, JS, Java, etc.)
- Language Selection: Dropdown or input selection
- Model Used: Transformer (Codex, CodeT5)
- Validation: Syntax checker, linter
- Interface: Text input box, code output area

Theme: AI for Creativity & Entertainment

The Style Weaver - Artistic Style Transfer

Objective:

Extract fashion trends and visual elements from social media using AI and apply style transfer for fashion branding.

Components:

- **Data Mining:** Use scraping tools to gather Instagram posts.
- **Image Analysis:** Use CNNs to identify colors, textures, styles.
- **Text Analysis:** NLP for captions, hashtags, emojis.
- **Generative AI:** Use GANs or VQGAN+CLIP to generate new fashion visuals.

Key Parameters:

- Input: Social media image + text (captions, hashtags)
- Output: Stylized image or visual trend analysis
- Image Processing: CNNs for color/style extraction
- Text Processing: NLP (hashtags, sentiment, keywords)
- Generative Models: GANs, VQGAN+CLIP
- Clustering: K-means, DBSCAN for fashion categories

Theme: AI Animation and Visualizations

The Motion Painter - AI for Animating Still Images

Objective:

Animate static images by adding realistic or stylistic motion effects.

Features:

- Upload image interface.
- Select motion zones (e.g., water, trees).

- Apply pre-trained models for motion synthesis.
- Export as video or GIF.

Key Parameters:

- Input: Still image
- Motion Selection: User-defined or auto-detected areas
- Output: Short animation (GIF/MP4)
- Motion Techniques: Optical flow, video GANs
- UI Tools: Drawing tool to mark animated regions
- Export Formats: MP4, GIF, WebP

The VFX Assistant - Simple Visual Effects Automation**Objective:**

Simplify visual effects editing using AI automation.

Features:

- Background removal without green screen.
- Object tracking for overlays.
- Basic rotoscoping (object isolation).

Key Parameters:

- Input: Video or image
- Output: Modified visuals (removed background, tracked object)
- Segmentation: DeepLabV3, U-Net
- Tracking: DeepSORT, OpenCV trackers
- Tools: Background removal, rotoscoping, motion overlay
- Object tracking (OpenCV, DeepSORT).

The Visual Explainer - Live Stream Object/Concept Recognition

Objective:

AI system that identifies key concepts/objects during live video streams.

Features:

- Real-time video analysis.
- Pop-up explanations or tooltips. Vocabulary builder for students.

Key Parameters:

- Input: Live video stream
 - Output: Labels or tooltips with identified concepts
 - Detection: YOLOv8, EfficientDet
 - Concept Mapping: Pre-defined ontology
 - Display: Real-time overlay with explanations
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Theme: AI in Cybersecurity

The Phishing Detective - Email Fraud Detection

Objective:

Detect and flag malicious emails to prevent phishing attacks.

Features:

- Scan subject lines, body text, links, and metadata.
- Assign risk scores.
- Visual warning dashboard for inboxes.

Key Parameters:

- Input: Email content, headers, metadata
- Output: Risk score, threat label (Phishing/Not)
- NLP Techniques: BERT for classification

- Dashboard: Risk visualization, email metadata

DATASETS:

The Code Conjurer – Natural Language to Code Snippets

CMU CoNaLa Dataset: Contains 2,379 training and 500 test examples of natural language intents paired with Python code snippets. [Papers with Code+1Papers with Code+1](#)

TinyCodes Dataset: Offers 1.6 million short code snippets across various programming languages, aiding in training models for code generation. [Hugging Face](#)

The Stack: A comprehensive dataset of source code from GitHub repositories, useful for pre-training code language models. [Hugging Face](#)

Theme: AI for Creativity & Entertainment

The Style Weaver – Artistic Style Transfer

Stylized Image Dataset: Ideal for training neural style transfer models, this dataset includes images with various artistic styles. [Kaggle+1GitHub+1](#)

Anime Style Transfer Dataset: Comprises 820 training and 93 test pairs of real faces and their anime-style representations, suitable for style transfer tasks. [Papers with Code+1GTS AI+1](#)

Theme: AI in Cybersecurity

The Phishing Detective – AI for Identifying Malicious Emails

Phishing Email Dataset (Kaggle): Comprises approximately 82,500 emails, with a balanced mix of spam and legitimate messages, aiding in training phishing detection models. [ISU Cose+3Kaggle+3UT Research Info+3](#)

Phishing Email Detection v2.0: Offers a comprehensive dataset combining email messages and URLs for phishing detection tasks. [Hugging Face](#)

FULL STACK DEVELOPMENT

Theme: The Future of Work & Learning

The Team Catalyst - Project Management Tool

Objective:

Enable better team collaboration through smart task tracking.

Key Parameters:

- Modules: Project creation, task assignment, chat, deadlines
- Users: Admins, team members
- Statuses: To-Do, In Progress, Done
- Timeline: Gantt chart, calendar
- Backend: API for CRUD operations
- Realtime: WebSockets or Firebase

Features:

- Create/manage projects.
- Assign tasks, set deadlines.
- Kanban boards, Gantt charts.
- Real-time chat and notifications.

The Learning Path - Personalized Learning Recommender

Objective:

Provide personalized learning paths using user profiles and goals.

Key Parameters:

- Input: User interests, skill level, goal
- Output: Ranked list of resources
- Recommendation Engine: Collaborative filtering, NLP ranking

- Feedback: Rating system to improve future suggestions

Features:

- User onboarding to set goals.
 - Recommender engine for courses/videos/articles.
 - Feedback system for improved suggestions.
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Theme: Facilitating Travel & Exploration

The Local Gem Finder - Hyperlocal Discovery

Objective:

Help users discover underrated locations and services based on hyperlocal data.

Features:

- Location-based recommendations.
- Community ratings and tips.
- Map-based interface.

Key Parameters:

- Input: User location, interests
 - Output: List/map of local spots
 - Data Sources: Crowd-sourced reviews, geo-tagging
 - AI Techniques: Clustering, recommendation systems
 - Interface: Map UI with search/filter
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Theme: Building Better Infrastructure & Services

The City Reporter - Community Issue Reporting

Objective:

Platform for citizens to report local issues and track resolutions.

Features:

- Upload image and location tag.
- Issue classification (e.g., pothole, lighting).
- Notify relevant municipal body.

Key Parameters:

- Input: Image, description, location
- Output: Issue ticket with tracking ID
- Classification: NLP categorization (pothole, light, etc.)
- Location Tagging: Google Maps API
- Authorities: Notification via email/API

Theme: Creating Engaging Digital Experiences**The Interactive Narrative - Branching Story Platform****Objective:**

Allow creators to write interactive stories and readers to explore narrative paths.

Features:

- Story builder with branching logic.
- Save user progress.
- Comment and share system.

Key Parameters:

- Story Structure: Tree/graph of decisions
- User Input: Choices that affect next story part
- Author Tools: Drag-and-drop story map builder
- Reader Features: Save progress, bookmarks

- Format: JSON for stories, HTML5/CSS for front-end rendering

